

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/01

Paper 1 Multiple Choice

15 September 2016

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

1 The electron micrograph of a cell is shown below.



Which statements are true?

- 1 P is made up of phospholipids and amino acids.
- 2 Q and R are found in all eukaryotic cells.
- 3 One of the processes in R synthesises ATP.
- 4 S contains only euchromatin but not heterochromatin.
- 5 Q, R and S contain both RNA polymerase and double stranded DNA.

A 1 and 3 only

B 3 and 5 only

C 1, 3 and 4 only

D 2, 3 and 5 only

2 During the production of apple juice, enzymes are used to break down the components of cell walls. What is the product of this hydrolysis?

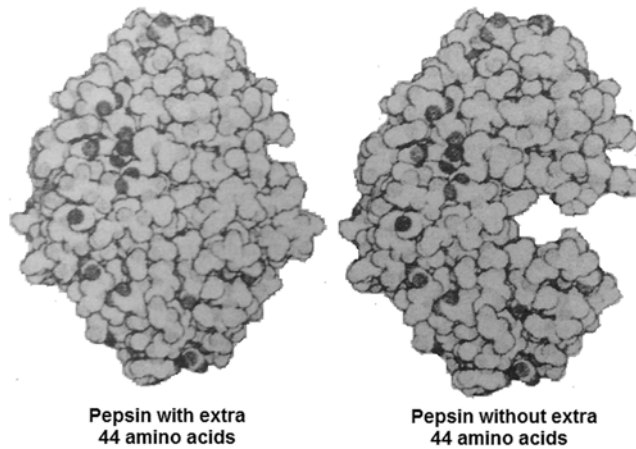
A amylose

B maltose

C α -glucose

D β -glucose

- 3 Protein-digesting enzymes are synthesised inside cells as inactive pro-enzymes, which are then activated outside the cell. Pepsin is synthesised with an extra 44 amino acids. This extra chain is removed in the stomach allowing the enzyme to digest proteins.



Which statement is incorrect?

- A There is a change in the tertiary structure of the protein molecule when the extra amino acids are removed.
- B Removal of the 44 amino acids is a hydrolysis reaction.
- C** The extra 44 amino acids serve as competitive inhibitor, preventing the binding of the substrate to the active site of the enzyme.
- D The change in shape of the pepsin upon activation is due to a change in the primary structure.

- 4 The diagram shows some *Allium sp.* plant cells in various stages of mitotic cell cycle labelled **W** – **Z**.

**W****X****Y****Z**

Which correctly describes the structure or behaviour of chromosomes at the various stages?

- A** The chromosomes at **W** are less condensed as compared to those at **Y**.
- B** The homologous chromosome pairs are separated to opposite poles of the cell during **X**.
- C** During **Y**, the chromosomes undergo semi-conservative DNA replication.
- D** The number of chromosomes per nucleus at **Z** is different from that at **W**.

5 During the replication of DNA in eukaryotic cells, some processes occur.

- 1 Free nucleotides are hydrogen bonded to those on the exposed strand.
- 2 Hydrogen are broken between the complementary base pairs.
- 3 The cell receives the signal to divide.
- 4 Covalent bonds form between adjacent nucleotides on the same strand.
- 5 The DNA double helix is unwound.

Which shows the correct order of some of the processes in DNA replication?

A 3 → 1 → 2 → 4

B 3 → 2 → 4 → 5

C 5 → 2 → 1 → 4

D 5 → 2 → 3 → 1

6 The following coding sequence is taken randomly from a bacterial genome.

3' – TTACGCTTCGAAATAGGAATATCATAGGCT-5'

Arg	CGU, CGC, CGA, CGG, AGA, AGG	Leu	UUA, UUG, CUU, CUC, CUA, CUG
Asp	GAU, GAC	Lys	AAA, AAG
Ile	AUU, AUC, AUA	Phe	UUU, UUC
Asn	AAU	Ala	GCG
Met	AUG	Ser	UCU, UCC, UCA, UCG, AGU, AGC
Glu	GAA	Tyr	UAU, UAC
Stop	UAG, UGA, UAA		

This sequence is cloned into a plasmid and transformed into a suitable host. What would be the first four amino acids of a peptide generated from this sequence as expressed by the host?

A Asn-Ala-Lys-Leu

B Met-Arg-Ser-Phe

C Met-Ile-Phe-Leu

D Leu-Arg-Phe-Glu

- 7 In the DNA sequence for sickle cell anaemia, adenine replaces thymine in a CTT triplet, forming the triplet CAT. During synthesis of the sickle cell haemoglobin molecule, the amino acid valine is incorporated instead of glutamic acid.

What is the anticodon in the transfer RNA molecule carrying this valine?

- A** CAU
- B CUA
- C GAU
- D GUA
- 8 Tay-Sachs disease is characterised by abnormal accumulation of lipid related compounds, which results in deterioration of cognitive and motor abilities.

It is caused by an autosomal recessive mutation in the allele coding for hexosaminidase A (HEXA), an enzyme that regulates the metabolism of phospholipids.

The base triplets in part of the coding DNA sequences for a normal HEXA allele and a mutant Tay-Sachs allele, as well as their corresponding amino acids are shown.

Normal HEXA allele	... CGT ...	ATA	TCC	TAT	GCC	CCT	GAC	...
	... Arg ...	Ile	Ser	Tyr	Gly	Pro	Asp	...
Tay-Sachs allele	... CGT ...	ATA	TCT	ATC	CTA	TGC	CCC	TGA ...
	... Arg ...	Ile	Ser	Ile	Leu	Cys	Pro	

Which combination correctly describes the nature of mutation that results in the Tay-Sachs allele?

	Changes to nucleotide sequences	Alteration of reading frame	Length of polypeptide
A	Deletion of 2 bases	Yes	Shorter
B	Insertion of 2 bases	Yes	Longer
C	Substitution of 4 bases	No	Unchanged
D	Insertion of 4 bases	Yes	Longer

9 Which features of viruses account for them being obligate parasites?

- 1 Nucleic acids in viruses can be cut into small fragments by bacterial restriction enzymes.
- 2 Each contains only one type of nucleic acid.
- 3 They can be crystallised.
- 4 They cannot synthesise ATP.
- 5 They have no cellular structure.
- 6 They have no enzymes involved in metabolism outside a host cell.

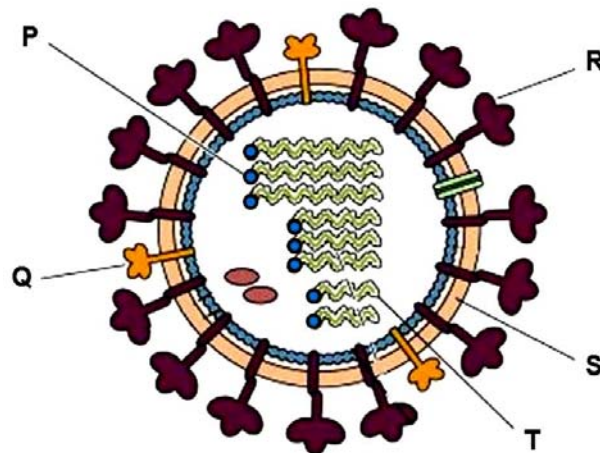
A 1, 3 and 5

B 1, 4 and 6

C 2, 3 and 5

D 2, 4 and 6

10 The diagram shows the structure of an influenza virus.



Which statements concerning the lettered components are correct?

- 1 Mutations that disrupt the function of **P** will result in the inability of the virus to initiate infection in the host cell.
- 2 Influenza viruses are able to cause pandemics in humans due to changes in **Q** and **R** by a slight mutation.
- 3 **Q** and **R** are unlikely targets for vaccination because they undergo mutation constantly.
- 4 The symptoms associated with influenza infection are directly due to loss of epithelial cells as the new viruses obtain **S** during budding.
- 5 The host cell enzymes are not required to form the complementary RNA from **T**.

- A** 1 and 5 only
- B** 1, 2 and 5 only
- C** 2, 3 and 4 only
- D** 3, 4 and 5 only

- 11** The bacterium, *Pneumococcus pneumonia*, form two types of colonies, whose cells are structurally different. Smooth (S) cells have thick outer capsules, but rough (R) cells lacks this capsule. S cells cause the disease pneumonia.

In 1928, Frederick Griffith found that:

- when R cells were mixed with heat-killed S cells and the mixture is injected into mice, some of the mice became infected and died
- living S cells with capsules could be isolated from these dead mice
- injection of heat-killed S cells alone or of living R cells alone did not cause disease in mice.

What can be concluded from these three observations to explain what happened when R cells were mixed with heat-killed cells?

- A** A heritable genetic change occurred in the R cells.
- B** R and S conjugated when mixed.
- C** R cells were changed into S cells by transduction.
- D** R cells were transformed by DNA from heat-killed cells.
- 12** A mutant strain of *E. coli* has been isolated in which the *lac* operon is not expressed in the presence of lactose. To understand the nature of this defect, this strain was mated so that it now contains an F factor containing a normal *lac* operon. The beta-galactosidase activities of both strains were compared in the presence and absence of lactose. The results obtained are shown in the table.

Strain	Addition of lactose	Activity of beta-galactosidase (percentage of parent strain)
Mutant	No	0
Mutant	Yes	0
Mated	No	0
Mated	Yes	100

Which part of the operon most likely has the mutation?

- A** *lacA*
- B** *lacI*
- C** promoter
- D** operator

13 The statements are about the different operon systems identified by Jacob and Monod in the 1950s.

- 1 In repressible systems, the unmodified regulator protein molecule always binds to DNA. In inducible systems, the unmodified regulator protein molecule never binds to DNA.
- 2 In repressible systems, the operon is active unless it is inhibited. In inducible systems, the operon is suppressed until it is activated.
- 3 Repressible systems are always regulated by negative control. Inducible systems are always regulated by positive control.
- 4 Repressible systems are usually regulated by the amount of end product. Inducible systems are usually regulated by the amount of substrate.

Which statements are true for both of these types of gene expression?

- A 1 and 4 only
- B 2 and 3 only
- C** 2 and 4 only
- D 2, 3 and 4 only

14 All the scenarios describe gene amplification except

- A during amphibian development, millions of additional rRNA genes exist as circular DNA in nucleoli separated from the chromosomes.
- B proto-oncogene replicates many copies of itself resulting in excessive amount of normal growth-stimulating protein.
- C** proto-oncogene duplicates itself and transposes to a new locus under new gene regulatory control system.
- D primers that flank a portion of DNA sequence, oligonucleotides, Taq polymerase are added to a reaction mixture and put through cycles of three different temperatures.

- 15 The electron micrograph shows part of a bacterial cell. The DNA molecule has been identified. Identify structures R and P as well as the process(es) that is(are) occurring.



	Structure P	Structure R	Process(es)
A	DNA polymerase	Daughter DNA	DNA replication
B	String of nucleosomes	mRNA	Transcription
C	Polyribosomes	Polypeptides	Translation
D	Polyribosomes	mRNA	Transcription and translation

- 16 Which combination of factors will affect the translational control of gene expression?

	Presence of carbohydrate group attached to polypeptide	Presence of repressor bound to 5' UTR of mRNA	Presence of ubiquitin attached to polypeptide
A	Yes	No	Yes
B	Yes	Yes	No
C	No	Yes	No
D	No	No	No

17 Cancer is a multi-step process. Some of the stages in cancer are listed.

- 1 Cell undergo uncontrolled cell division by mitosis.
- 2 Formation of tumour.
- 3 Mutations in tumour suppressor genes and proto-oncogenes.
- 4 Cells from the primary tumour metastasize to other sites via the bloodstream.

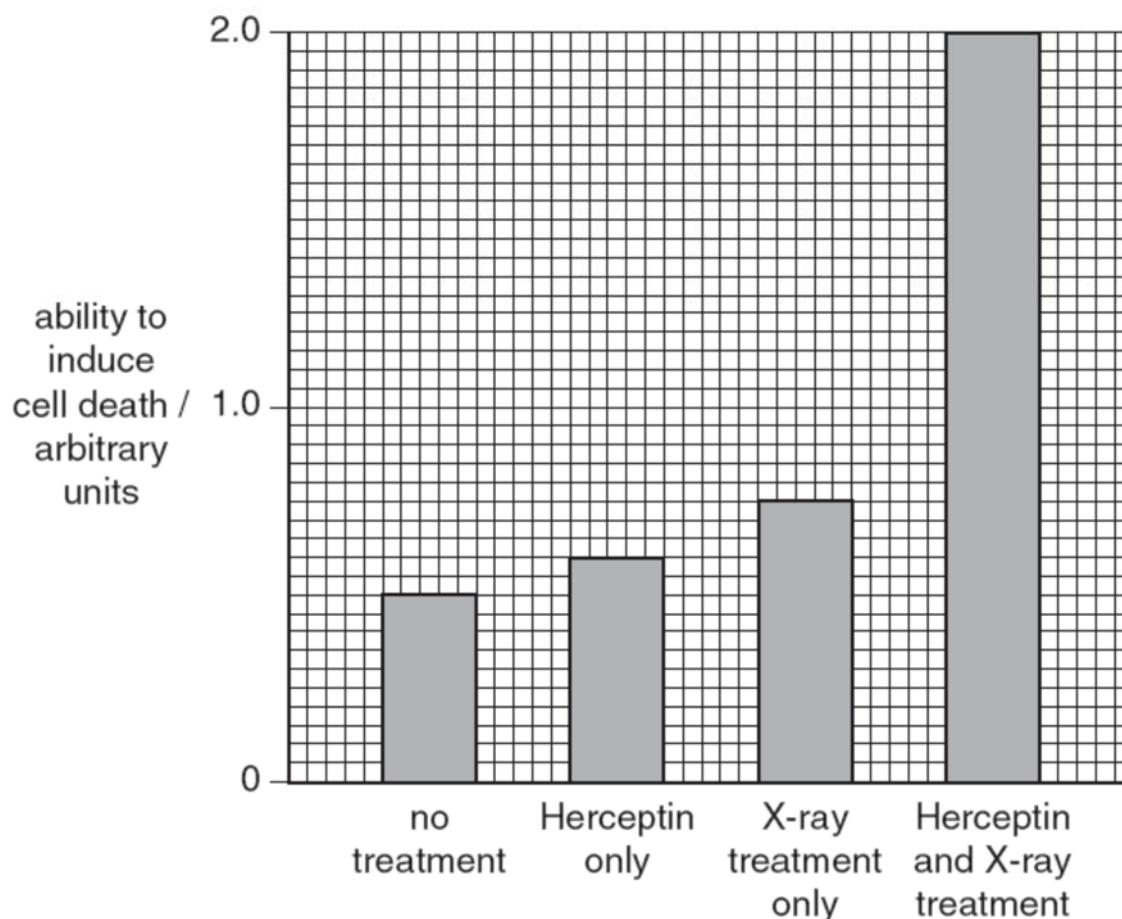
What is the correct sequence of events?

- A** 1 → 4 → 3 → 2
- B** 1 → 2 → 3 → 4
- C** 3 → 1 → 2 → 4
- D** 3 → 2 → 1 → 4

- 18 Herceptin is a monoclonal antibody used in the treatment of some breast cancers. It binds strongly to molecules of a receptor protein, HER2 that is produced in abnormally large quantities in the plasma (cell surface) membranes of about 30% of human breast cancers.

One experiment investigated the ability of different treatments to induce cell death in breast cancer cells.

Herceptin and X-ray treatment were used both separately and together. The results are shown.



Which is a valid conclusion of this study?

- A** The percentage increase in inducing cell death using Herceptin and X-ray compared with using Herceptin only is 233%.
- B** Herceptin is not effective in inducing cell death.
- C** X-ray cannot effectively induce cell death because it is a carcinogen.
- D** Binding of Herceptin to receptor proteins found in breast cancer cells prevents stimulation of uncontrolled cell division.

percentage increase in the ability to induce cell death of using Herceptin and X-ray treatment compared with using Herceptin only = $2.0 - 0.6 / 0.6 \times 100 \% = 233 \%$

- 19 In mice there are several alleles of the gene that controls the intensity of pigmentation of the fur.

The alleles are listed below in order of dominance with **C** as the most dominant.

C = full colour
C^{ch} = chinchilla
C^h = himalayan
C^p = platinum
C^a = albino

The gene for eye colour has two alleles. The allele for black eyes, **B**, is dominant, while the allele for red eyes, **b**, is recessive.

A mouse with full colour and black eyes was crossed with a himalayan mouse with black eyes. One of the offspring was albino with red eyes.

What is the proportion of offspring that will not resemble both parents?

- A 1/4
 - B 1/2
 - C** 7/16
 - D 9/16
- 20 A tall plant is crossed with a short plant, and the progeny are all intermediate in size between the two parental plants. The intermediate F_1 progeny were allowed to self-pollinate, and the F_2 progeny were also intermediate in size, but following a normal distribution. The results of the F_2 progeny is best explained by
- A environmental factors.
 - B multiple alleles.
 - C** additive effect of polygenes.
 - D epistasis.

- 21 In a series of breeding experiments using *Drosophila*, a linkage group composed of five genes was found to show the following recombination frequencies per 100 fertilised eggs.

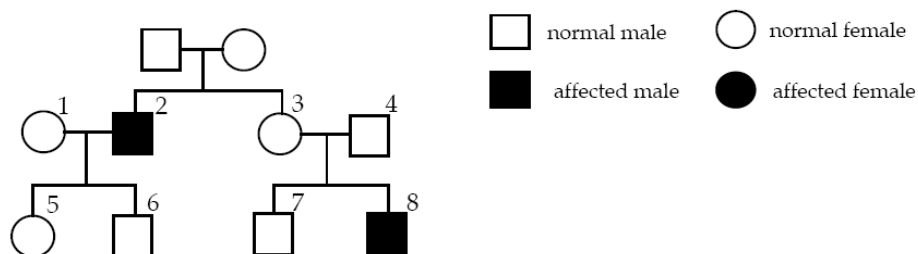
	Single bristles (<i>sn</i>)	Vermilion eye (<i>v</i>)	White eye (<i>w</i>)	Ruby eye (<i>rb</i>)	Tan body (<i>t</i>)
Single bristles (<i>sn</i>)	0.0	12.0	19.5	13.5	6.5
Vermilion eye (<i>v</i>)	12.0	0.0	31.5	25.5	5.5
White eye (<i>w</i>)	19.5	31.5	0.0	6.0	26.0
Ruby eye (<i>rb</i>)	13.5	25.5	6.0	0.0	20.0
Tan body (<i>t</i>)	6.5	5.5	26.0	20.0	0.0

From this table of results, it can be deduced that the correct sequence of the genes on the chromosome is

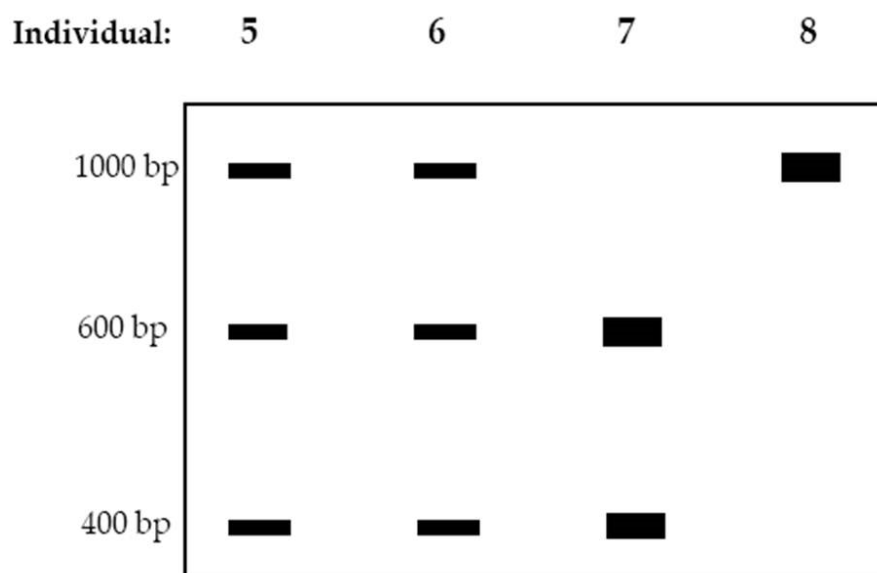
- A $sn - t - v - rb - w$
- B $sn - v - w - rb - t$
- C** $w - rb - sn - t - v$
- D $w - rb - t - sn - v$

- 22 A study on a genetic condition was conducted. The mutant allele differs from the wild-type allele by a single base-pair (bp) substitution. This substitution eliminates a restriction site of *EcoRI* that is present in the wild-type allele.

A pedigree of a family exhibiting this condition is shown.



Researchers isolated DNA from four individuals in the pedigree. Using PCR techniques, they amplified a 1000 bp portion of their DNA that includes the site affected by the mutation. They digested the PCR products with *EcoRI* and analysed the resulting DNA fragments on a gel.



Based on the data, the gene is located on

- A X chromosome.
- B both chromosomes.
- C **autosome.**
- D Y chromosome.

23 Test crossing some *Drosophila* gave the following results:

Red eye, normal wing	58
Purple eye, vestigial wing	42
Red eye, vestigial wing	38
Purple eye, normal wing	62

The ratio of phenotypes expected for such a dihybrid test cross is 1:1:1:1.

A χ^2 test was performed to test the significance of the difference between the observed and the expected results.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

Where Σ = 'sum of...'
 v = degrees of freedom
 c = number of classes
 O = observed 'value'
 E = expected 'value'

degrees of freedom	probability, p						
	0.50	0.20	0.10	0.05	0.02	0.01	0.001
1	0.46	1.64	2.71	3.84	5.41	6.64	10.38
2	1.39	3.22	4.61	5.99	7.82	9.21	13.82
3	2.37	4.64	6.25	7.82	9.84	11.35	16.27
4	3.36	5.59	7.78	9.49	11.67	13.28	18.47

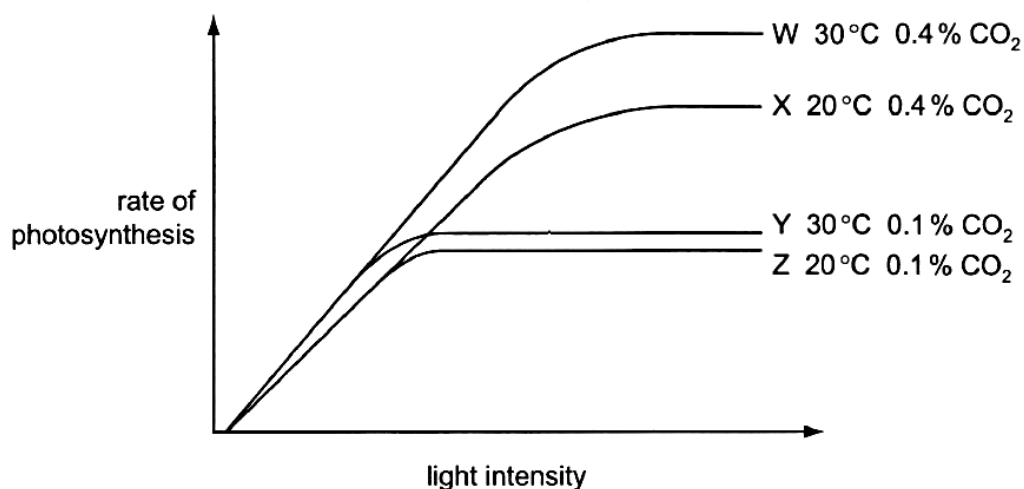
Which combination correctly describes the result of the χ^2 test?

	Probability	Significance of difference between observed and expected results	Loci of genes
A	< 0.05	not significant	different chromosomes
B	< 0.05	significant	same chromosomes
C	> 0.05	not significant	different chromosome
D	> 0.05	significant	same chromosome

24 Some photosynthetic organisms contain chloroplasts that lack photosystem II that are able to survive. The best way to detect the lack of photosystem II in these organisms would be

- A to test for carbon fixation in the dark.
- B to test for liberation of oxygen in the light.
- C to test for presence of starch.
- D to determine if they have thylakoids in the chloroplasts.

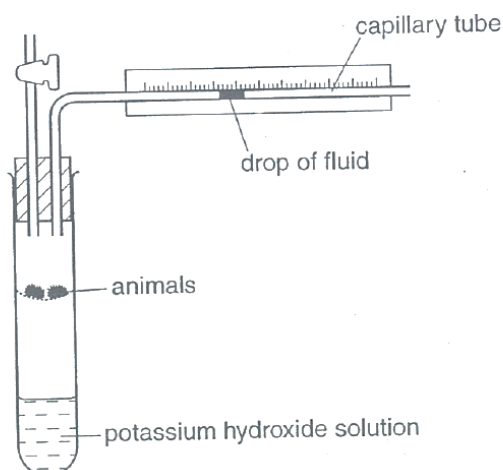
- 25 The diagram shows the results of an investigation into the effect of changing light intensity on the rate of photosynthesis at two different carbon dioxide concentrations and two different temperatures.



Which factor is limiting the rate of photosynthesis shown on curve **X** at high light intensities and which curve supports this?

- A** Carbon dioxide concentration, curve **Y** supports as a decrease in carbon dioxide concentration decreases the photosynthetic rate.
- B** Carbon dioxide concentration, curve **Z** supports as the photosynthetic rate becomes constant at lower light intensities.
- C** Temperature, curve **W** supports as an increase in temperature increases the photosynthetic rate.
- D** Temperature, curve **Z** supports as the photosynthetic rate becomes constant at lower light intensities.

26 A respirometer was set up to investigate the rate of respiration as shown below.

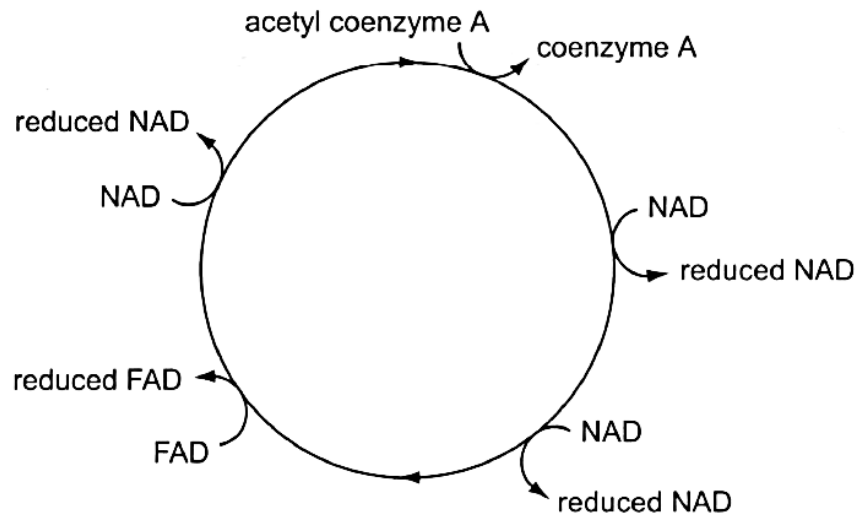


At the end of the experiment, it was found that the drop of fluid significantly moved to the left more than expected. Mitochondria from the animals were isolated to investigate the cause.

Which is a possible observation made about the mitochondria which best explains the results?

- A Cytochrome *b* in the electron transport chain was non-functional.
- B** Higher than normal amount of cristae observed.
- C Proton pump is decoupled from the electron carriers.
- D ATP synthase was not-functional.

27 The diagram shows the reactions of the hydrogen carriers in the Krebs cycle.

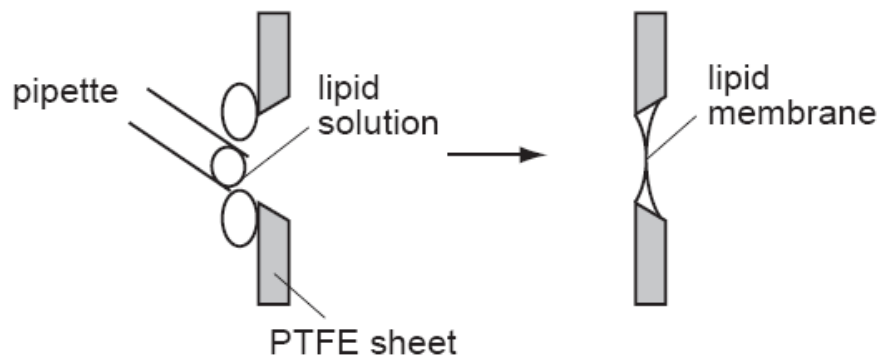


The average yield of ATP, in oxidative phosphorylation, is 3 molecules from each molecule of reduced NAD and 2 molecules from each molecule of reduced FAD.

What is the average yield of ATP, in oxidative phosphorylation, from the hydrogen carriers reduced in the Krebs cycle, from one molecule of glucose?

- A 11
- B 22**
- C 34
- D 36

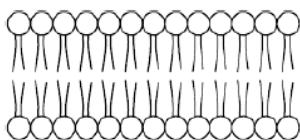
- 28 Lipid membranes can be formed in the laboratory by painting phospholipids over a PTFE sheet with a hole in it. Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+ .



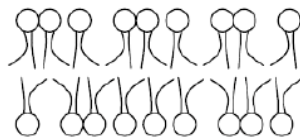
In one experiment with Na^+ ions, no current flowed across the membrane until a substance called gramicidin was added which allows the current to flow.

What is the characteristics of gramicidin?

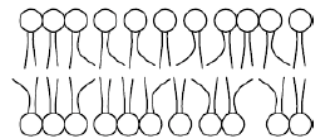
- A a carbohydrate molecule found only on the outside of the membrane
 - B a non-polar lipid which passes all the way through the membrane
 - C** a protein molecule with both hydrophilic and hydrophobic regions
 - D a protein molecule which has only hydrophobic regions
- 29 A mutant strain of bacterium, *Escherichia coli* was incapable of incorporating unsaturated phospholipids into its plasma membrane. Which correctly depicts and describes the membrane of such bacteria?



Type I



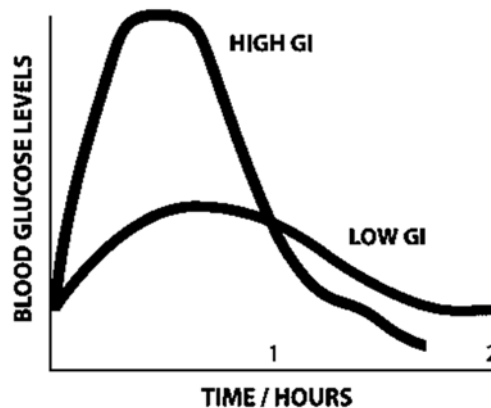
Type II



Type III

- A The membrane would appear as Type I and would be more fluid at low temperatures.
- B** The membrane would appear as Type I and would be less fluid at low temperatures.
- C The membrane would appear as Type II and would be more fluid at low temperatures.
- D The membrane would appear as Type III and would be no different from normal bacterial membranes.

- 30** Glycemic Index (GI) is a measurement on carbohydrate-containing foods and their impact on the blood sugar. The graphs show the impact on blood glucose levels of eating high GI food as compared to low GI food.



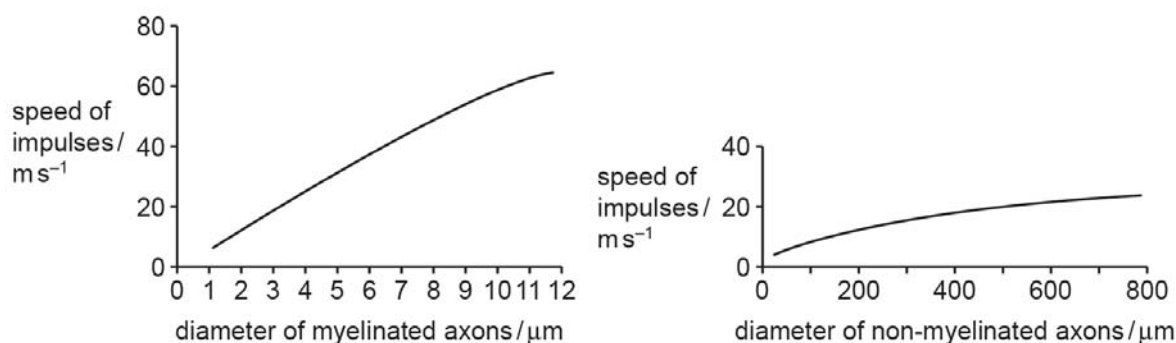
Which is the most likely response to eating high and low GI foods?

- A** Level of insulin released by eating high GI food is higher than eating low GI resulting in more fat cells taking in glucose.
 - B** Level of insulin released by eating high GI food is the same as eating low GI food but takes longer time for blood glucose to reach set point.
 - C** Level of glucagon released by eating high GI food is lower than eating low GI food resulting in more glucose stored as glycogen in the muscle.
 - D** Level of glucagon released by eating high GI food is the same as level of insulin released as both works antagonistically to bring blood glucose back to set point.
- 31** Which of the following process(es) lead(s) to signal amplification in the cell?
- 1 binding of multiple signal molecules to the membrane bound receptor
 - 2 phosphorylation of the first kinase in a phosphorylation cascade
 - 3 Branching of different signaling pathways that produces multiple cellular responses
 - 4 Activation of numerous intracellular receptors that acts as transcription factors in gene expression
- A** 2 only
 - B** 1 and 4 only
 - C** 2 and 3 only
 - D** All of the above

- 32 The ability of organisms to respond rapidly to stimuli is limited by the speed of the impulses in their neurons.

The axons of invertebrate neurones lack a myelin sheath. The axons of most vertebrate neurones are myelinated.

The graphs show the speed of impulses in these two types of axon.



Which statements about these data are correct?

- 1 The action potential in myelinated axons is greater than the action potential in non-myelinated axons.
- 2 The speed of impulses is changed by the diameter of the axon.
- 3 Increasing the diameter of a myelinated axon causes a greater change to the conduction speed than increasing the diameter of a non-myelinated axon.
- 4 The presence of myelin increases the speed at which impulses are conducted.

A 1, 2 and 3

B 1, 2 and 4

C 1, 3 and 4

D 2, 3 and 4

- 33** Red light can only penetrate about 10m through water, but blue and green wavelengths can penetrate more than twice that distance.

Pigments which can absorb light of one colour and use the energy to emit light of another colour are called 'fluorescent'. Many reef corals living at depths between 10m and 20m fluoresce red.

Some fish living at depths between 10m and 20m also have pigments which fluoresce red. These pigments often surround the eyes and form patches on the body and fins, in patterns unique to each species.

Which of these suggested reasons could explain the selective advantage of fluorescence for these fish?

- 1 Fluorescent patches provide camouflage.
- 2 Recognition of their own species avoids interbreeding with other species.
- 3 The fluorescent patches do not attract predators more than 10m away.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

- 34** Compared to allopatric speciation, sympatric speciation develops reproductive isolation without a

A barrier to mating.

B geographical barrier.

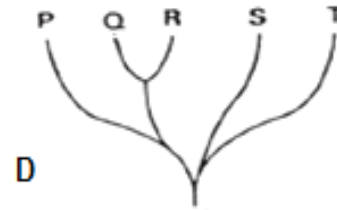
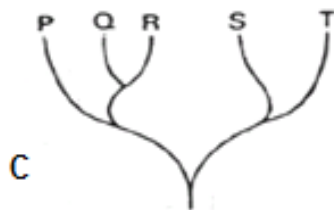
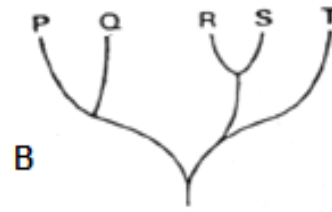
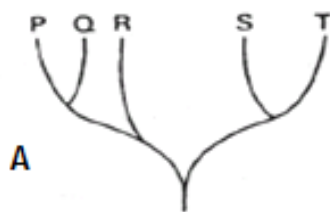
C barrier to gene flow.

D genetic change.

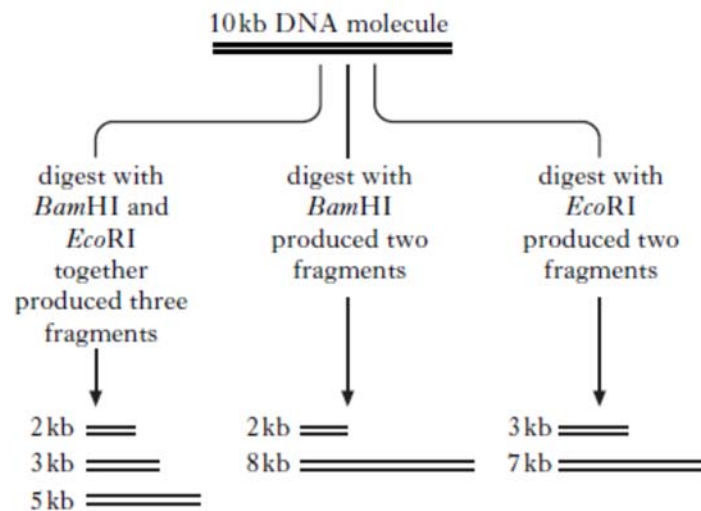
- 35 Five species, P, Q, R, S and T, possess the protein cytochrome c oxidase. The primary structure of the protein was determined for each species and the number of amino acid differences between species is given below.

	P	Q	R	S	T
P	0	7	8	20	22
Q	7	0	3	19	17
R	8	3	0	18	21
S	20	19	18	0	10
T	22	17	21	10	0

From the evidence, which phylogenetic tree represents the probable evolutionary relationship between the species? **C**



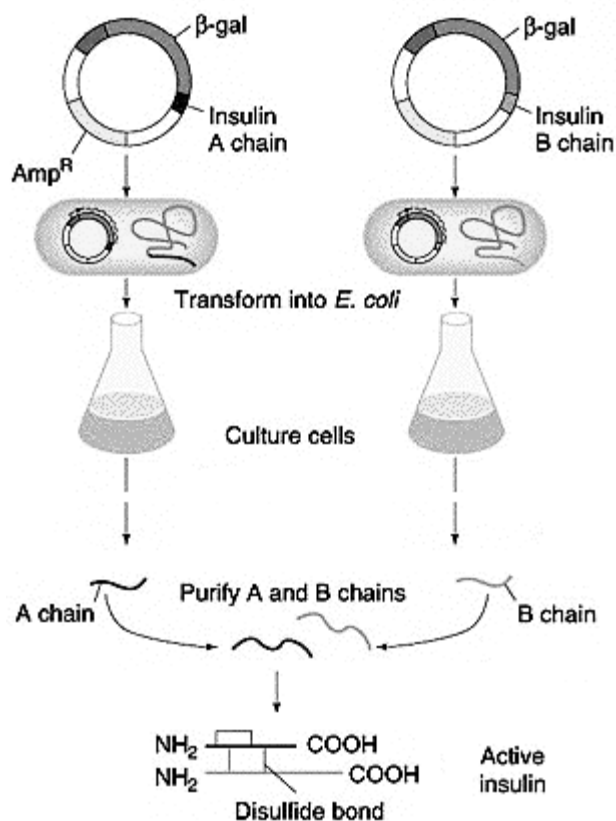
- 36 A piece of DNA was digested using the restriction enzymes *Bam*HI and *Eco*RI. The results are shown below.



Which is the correct restriction map that can be drawn from these results?

- A**
- B**
- C**
- D**

- 37 The insulin hormone is made up of two different polypeptide chains, A-chain and B-chain, joined together by disulphide bonds. The human insulin gene codes for both polypeptide chains. To produce the insulin hormone using *E. coli*, cDNA coding for each polypeptide chain are inserted into separate plasmids and used to transform separate batches of bacteria. The synthesised polypeptide chains are then mixed together to form the hormone.



Which is a valid explanation for producing the two chains separately?

- A Bacterial RNA polymerases cannot recognise eukaryotic promoters.
- B** Bacterial cells are not able to carry out post-translational modification of insulin.
- C Bacterial cells are unable to excise introns from eukaryotic RNA.
- D Bacterial ribosomes can only translate polycistronic mRNA.

38 Some of the goals of the Human Genome Project are:

- to determine the sequence of the entire human genome
- to identify all the genes in the human genome
- to find the locus of all the genes on the 46 human chromosomes.

What are some ethical concerns arising from the goals stated?

- 1 Anthropologist tracing the ancestry of human populations.
- 2 Parents choosing embryos for implantation only after tests for acceptable genes.
- 3 Insurance company offering cheaper rates to people with genetic disposition to fewer diseases.
- 4 Scientists developing tests for only some disease-causing genes.
- 5 Genetic counselors giving advice to people who are genetically pre-disposed to risks.

A 2 and 3

B 3 and 4

C 1 and 5

D 4 and 5

39 Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine and the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a blood cell.

Which statement explains this?

A Different stem cells have only the genes required for their particular cell line.

B Genes not required for the differentiation of a particular cell line are methylated.

C Binding of repressor molecules prevent the expression of genes not required for a particular cell line.

D Expression of genes not required for a particular cell line is controlled at translational level.

40 Retrovirus is a more suitable vector than liposome for gene therapy as it

- A** is non-pathogenic and does not trigger an immune response.
- B** causes mutation to the defective allele making it an effective form of treatment.
- C** is more specific to the target cells due to the presence of phospholipid bilayer that can fuse with the target cell.
- D** results in stable gene integration resulting in the therapy being more long lasting.